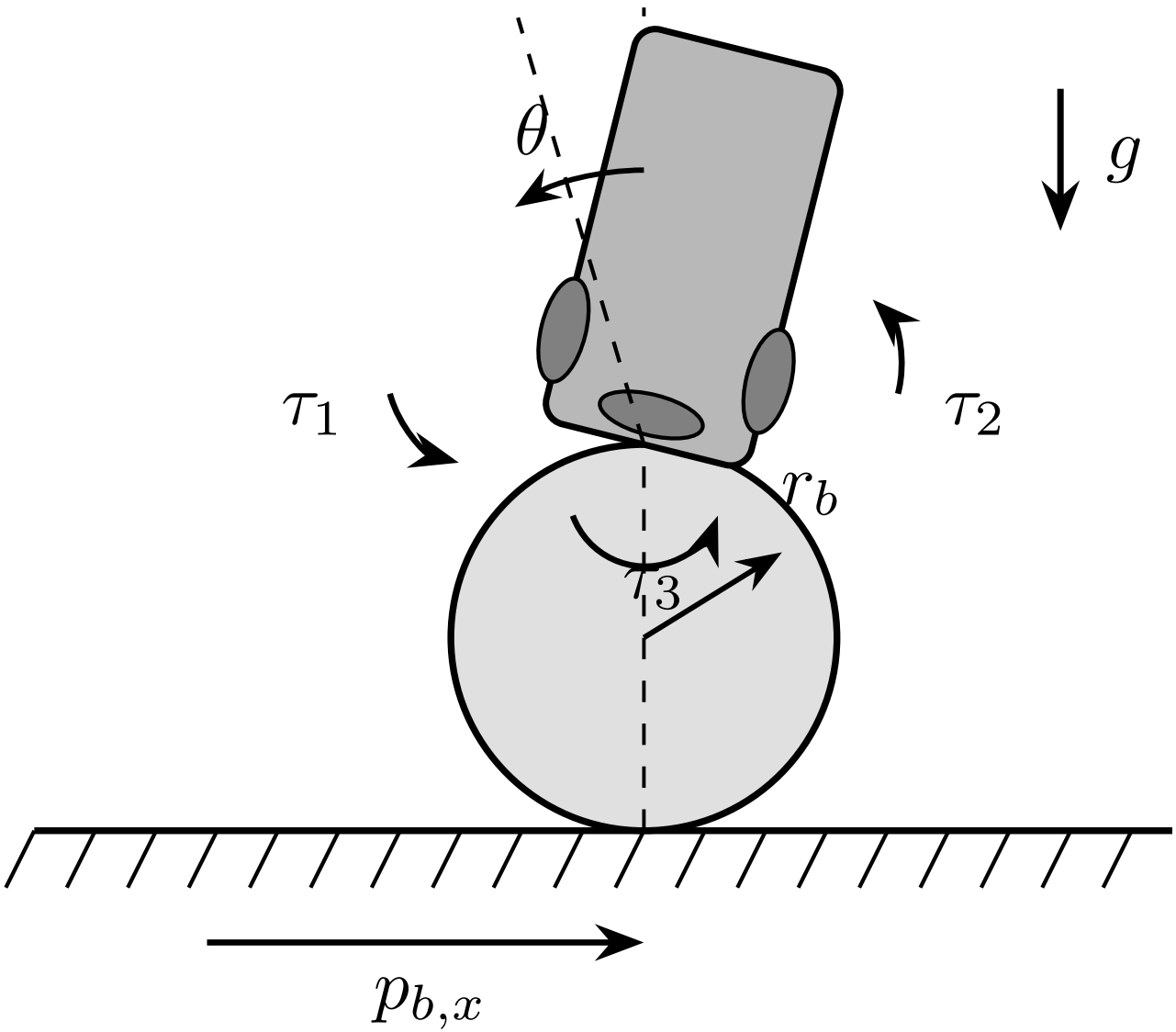

$$q = \begin{bmatrix} p_{b,x} \\ p_{b,y} \\ \theta_z \\ \theta_y \\ \theta_x \end{bmatrix}$$

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix} \in \mathbb{R}^{10}$$

$$u = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$$